

EMCH 260

SOLID MECHANICS

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Axial Load

4

Chapter 4.2

- ✓ Understand the statically indeterminate axially loaded members
- ✓ Deal with compatibility conditions
- ✓ Solve the statically indeterminate axially loaded problems

DETERMINATE VS. INDETERMINATE

- For a bar fixed-supported at one end, equilibrium equations is sufficient to find the reaction at the support. Such a problem is **statically determinate**.

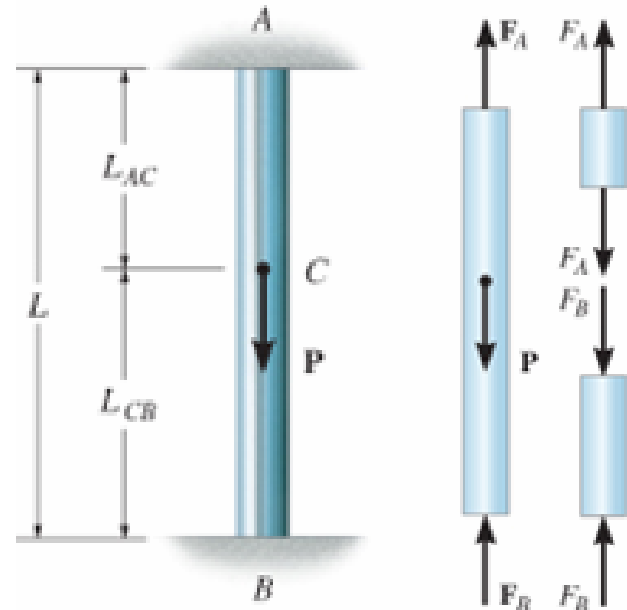


- If bar is fixed at both ends, then two unknown axial reactions occur, and the bar is **statically indeterminate**.

$$+\uparrow \sum F = 0 \rightarrow$$

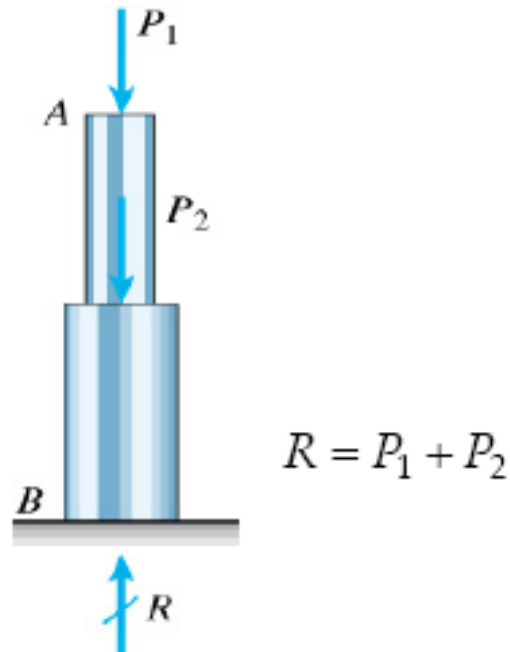
$$F_B + F_A - P = 0 \quad \text{Eq. (1)}$$

One equilibrium equation is not sufficient to determine both reactions on the bar



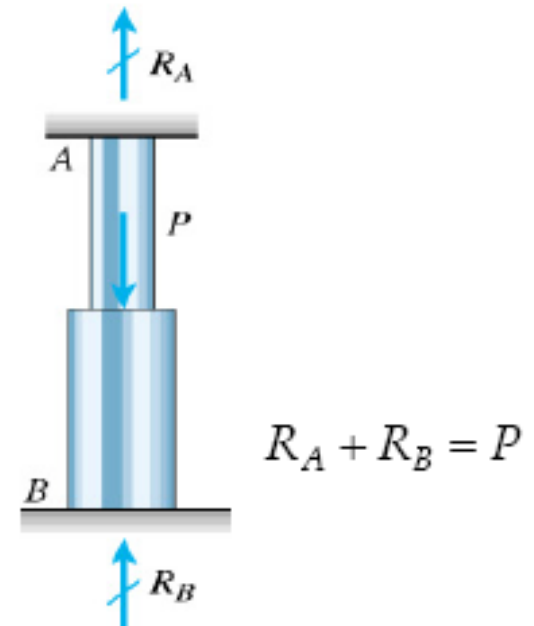
DETERMINATE VS. INDETERMINATE (CONT'D)

Statically Indeterminate Structures



Statically Determinate Structures

Reactions and internal forces can be determined solely from free-body diagrams and *equations of equilibrium* without knowing the properties of the materials.



Statically Indeterminate Structures

In addition to the *equilibrium consideration*, equations pertaining to the displacement of the structure, called *equations of compatibility*, and the *relations between forces and displacements* are usually needed.

STATICALLY INDETERMINATE AXIALLY LOADED MEMBER

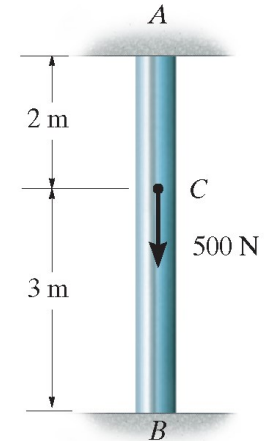
- To establish additional equation, one should consider geometry of deformation. Such an equation is referred to as a **compatibility or kinematic condition**.
- Since relative displacement of one end of bar to the other end is equal to zero, since end supports fixed,

$$\delta_{A/B} = 0$$

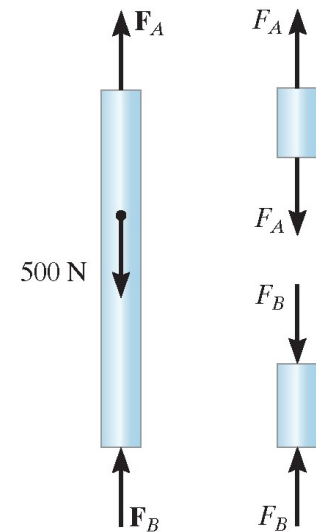
This equation can be expressed in terms of applied loads using a *load-displacement relationship* ($\delta = NL/(AE)$), which depends on the material behavior.

$$\delta_{AC} = \frac{F_A L_{AC}}{AE} \quad \delta_{CB} = -\frac{F_B L_{CB}}{AE}$$

$$\delta_{A/B} = \delta_{AC} + \delta_{CB} = 0 \quad \text{Eq. (2)}$$



(a)



(b)



(c)

PROCEDURE FOR ANALYSIS

Step 1: Equilibrium (exactly same as what we did before)

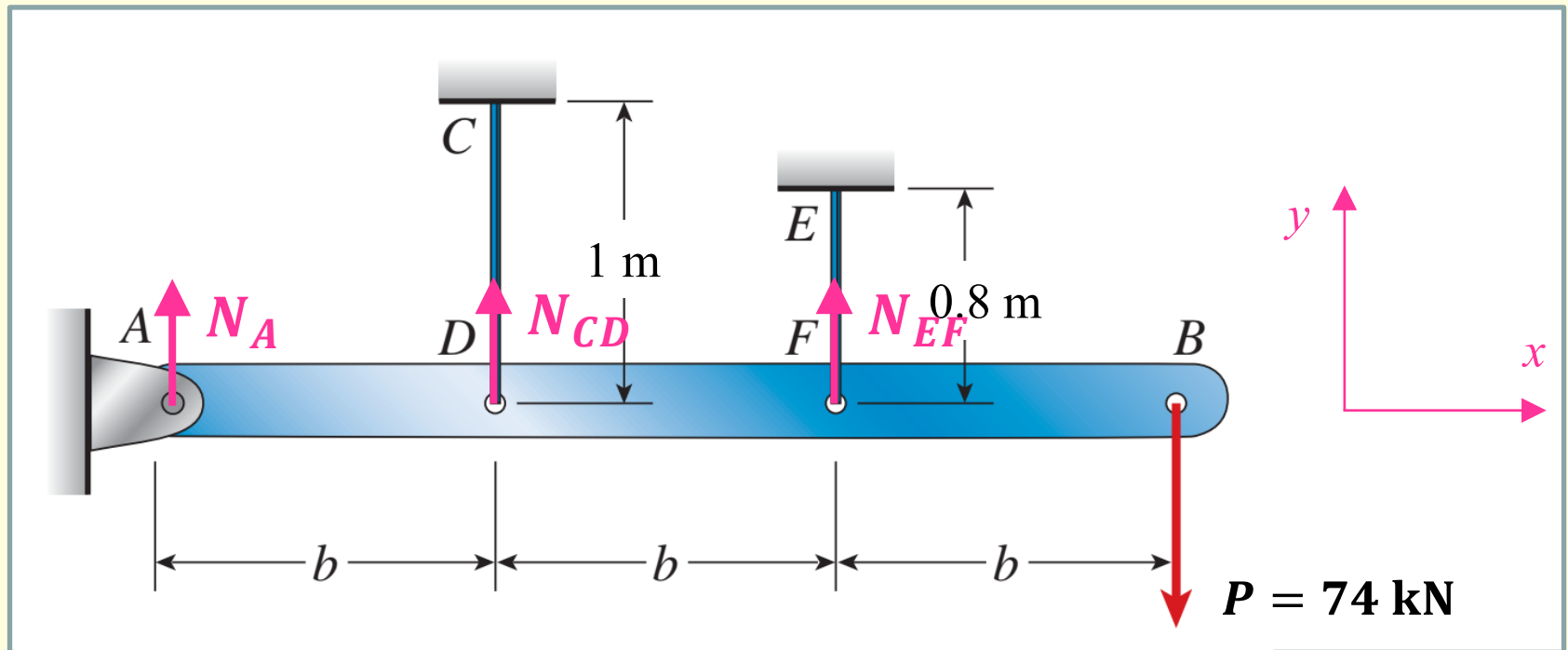
- Draw a free-body diagram of member to identify all forces acting on it
- If unknown reactions on free-body diagram greater than number of equilibrium equations, then problem is statically indeterminate
- Write the equations of equilibrium for the member

Step 2: Compatibility

- Draw a diagram to investigate elongation or contraction of loaded member
- Express compatibility conditions in terms of **displacements** caused by forces or loadings
- Use load-displacement relations ($\delta = NL/(AE)$) to relate unknown displacements to reactions (sometimes geometry conditions apply)
- Solve the equations. If result is negative, this means the force acts in opposite direction of that indicated on free-body diagram.

IN-CLASS EXAMPLE

A horizontal rigid bar AB is pinned at end A and supported by two wires (CD and EF) at points D and F . A vertical load $P = 74$ kN acts at end B of the bar. The bar has length 3 m and wires CD and EF have lengths of 1 m and 0.8 m, respectively. Also, wire CD has diameter of 10 mm and modulus of elasticity of 200 GPa; wire EF has diameter of 16 mm and modulus of elasticity of 100 GPa. Determine the displacement of the end B .



IN-CLASS EXAMPLE (cont'd)

Solutions

Step 1: Equilibrium

Assume the resultant forces in the wires and apply free body diagram to the rigid bar:

$$\sum F_y = 0 \quad \rightarrow \quad N_{CD} + N_{EF} + N_A = 74 \text{ kN} \quad (1)$$

$$\sum M_A = 0 \quad \rightarrow \quad N_{CD} + 2N_{EF} = 74 \times 3 = 222 \text{ kN} \cdot \text{m} \quad (2)$$

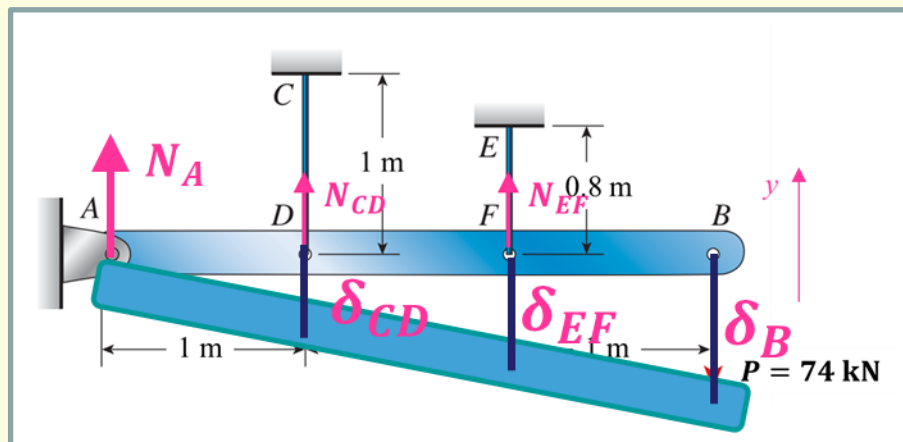
Step 2: Compatibility Condition

Referring to the geometry of the deformation of the two wires:

$$\delta_{EF} = 2\delta_{CD} \quad (3)$$

$$\delta_{EF} = \frac{N_{EF}L_{EF}}{E_{EF}A_{EF}}$$

$$\delta_{CD} = \frac{N_{CD}L_{CD}}{E_{CD}A_{CD}}$$



IN-CLASS EXAMPLE (cont'd)

Solutions, cont'd

Substituting these equations into Eq. (3) we get:

$$\frac{N_{EF}L_{EF}}{E_{EF}A_{EF}} = 2 \frac{N_{CD}L_{CD}}{E_{CD}A_{CD}}$$

$$\frac{N_{EF} \times 0.8}{100 \times 10^9 \times \frac{\pi}{4} \times 0.016^2} = 2 \times \frac{N_{CD} \times 1}{200 \times 10^9 \times \frac{\pi}{4} \times 0.01^2}$$

$$N_{CD} = 0.3125N_{EF} \quad (4)$$

Substituting Eq. (4) into Eq. (2) yields:

$$N_{EF} = 96 \text{ kN}$$

Finally, according to the geometry, the displacement of point B is

$$\delta_B = \frac{3}{2} \delta_{EF} = \frac{3}{2} \frac{N_{EF}L_{EF}}{E_{EF}A_{EF}} = 1.5 \times \frac{96000 \times 0.8}{100 \times 10^9 \times \frac{\pi}{4} \times 0.016^2} = 0.00573 \text{ m} \quad (\text{Ans.})$$

RELEVANT PRACTICE

- **Assignment 2 on Blackboard, problem #2-3 (go to Blackboard under the “Assignments” folder, **Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**
- **Optional Extra Exercise 2 on Blackboard, problem #1-2 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 140 – 146**
 - **In particular, Examples 4.5 – 4.7 (pp. 143 – 145)**